

BIOLOGY PAPER 1

SECTION B: Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) Present your answers in paragraphs wherever appropriate.
- (7) The diagrams in this section are **NOT** necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Please stick the barcode label here.

Candidate Number

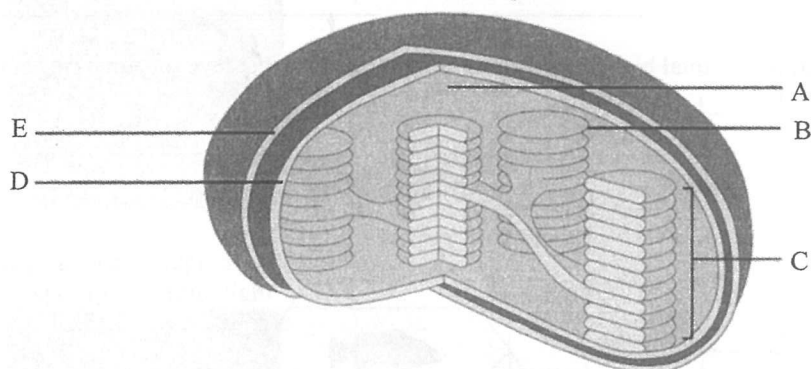
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SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

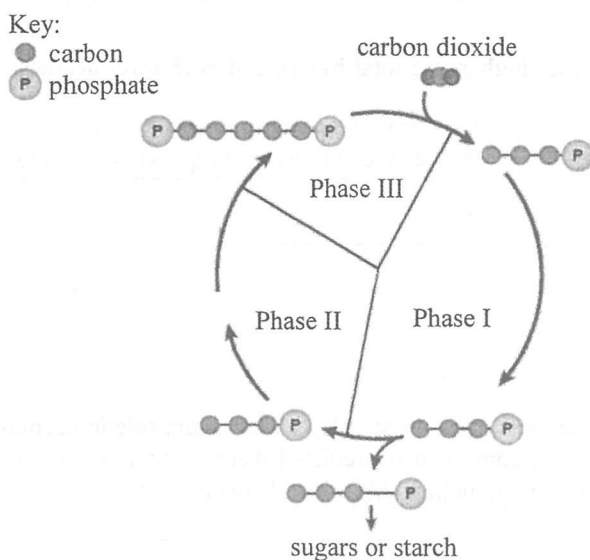
1. The diagram below shows the structures in a chloroplast:



- (a) Using the letters from the diagram, list **all** of the structures that contain photosynthetic pigments. (1 mark)

- (b) Structure C produces intermediates that are used in the Calvin Cycle. State the intermediates. (1 mark)

- (c) The diagram below shows a simplified Calvin Cycle:



Match the three phases with the following reactions: (2 marks)

Reactions

Regeneration of carbon dioxide acceptor

Reduction of 3C compound

Carbon dioxide fixation

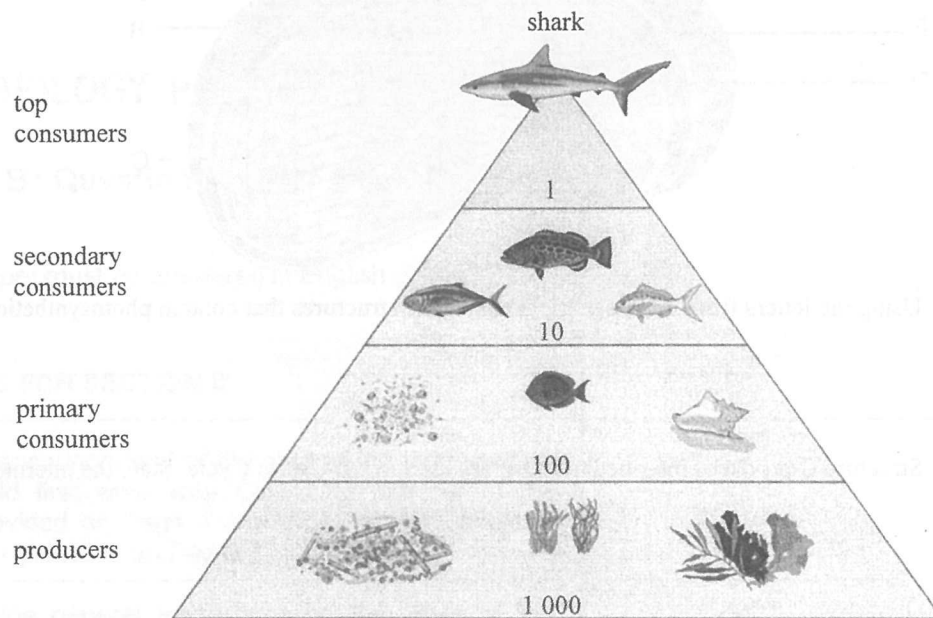
Phase

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2. The diagram below shows the total biomass (arbitrary unit) at each trophic level in a marine ecosystem:



- (a) As the trophic level becomes higher, the total biomass of each level decreases. Give *two* reasons for this phenomenon. (2 marks)

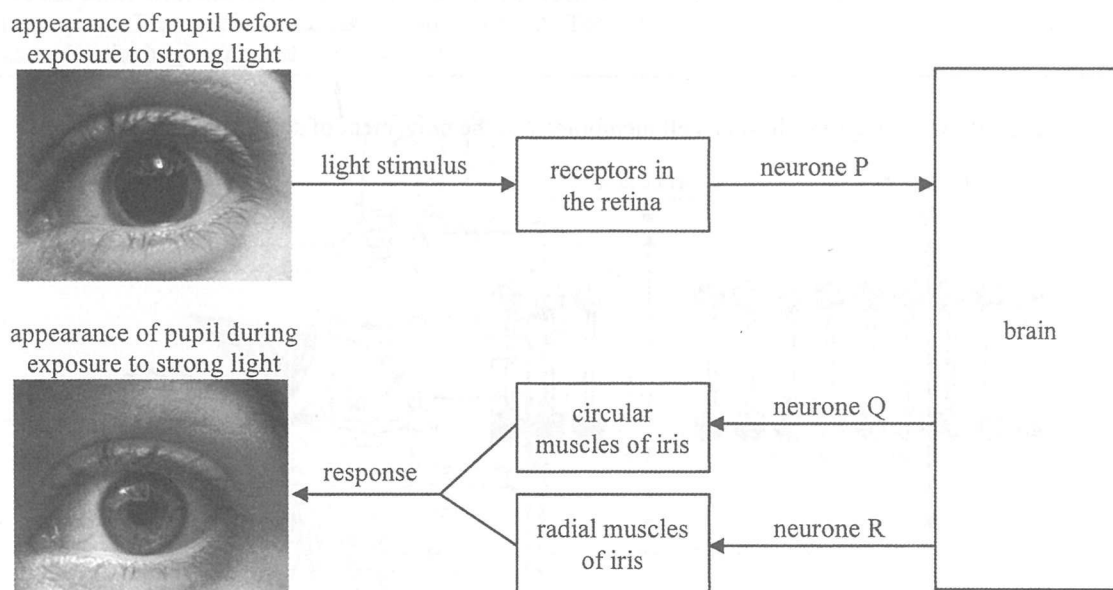
Reason 1:

Reason 2:

- (b) Sharks, being the top consumers in the ocean, play an important role in keeping the populations of other marine organisms under control. It is predicted that the extinction of sharks would result in overpopulation of producers. Explain how this would happen. (3 marks)

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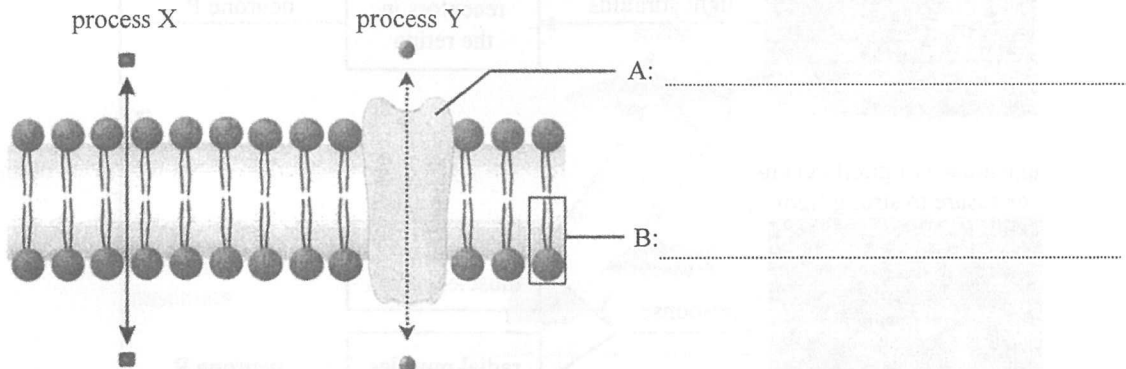
3. The diagram below shows a neural pathway involved in the coordination of pupil size when a human eye is exposed to strong light:



- (a) With reference to the above neural pathway,
- (i) state the **two** types of receptors located in the retina. (1 mark)
-
- (ii) state the type of neurone represented by Q and R. (1 mark)
-
- (b) Describe how the two sets of iris muscles work together to bring about the change in the pupil size shown above. State the significance of this response. (4 marks)
-
-
-
-
-
-
-
- (c) If someone falls unconscious, the response of the above neural pathway will be assessed to confirm if this pathway is still functioning. What does this assessment tell you about the nature of this neural coordination? (1 mark)
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4. The following diagram shows a cell membrane and the movement of some substances across it:



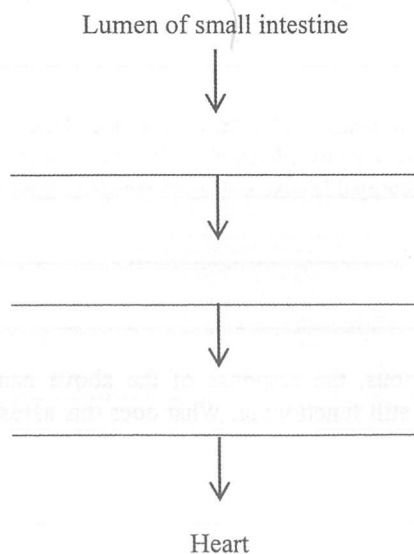
- (a) Label structures A and B. (2 marks)

- (b) X and Y represent two different processes by which substances pass through the cell membrane.

- (i) Give **one** digested product which is absorbed through process X in the human body. (1 mark)

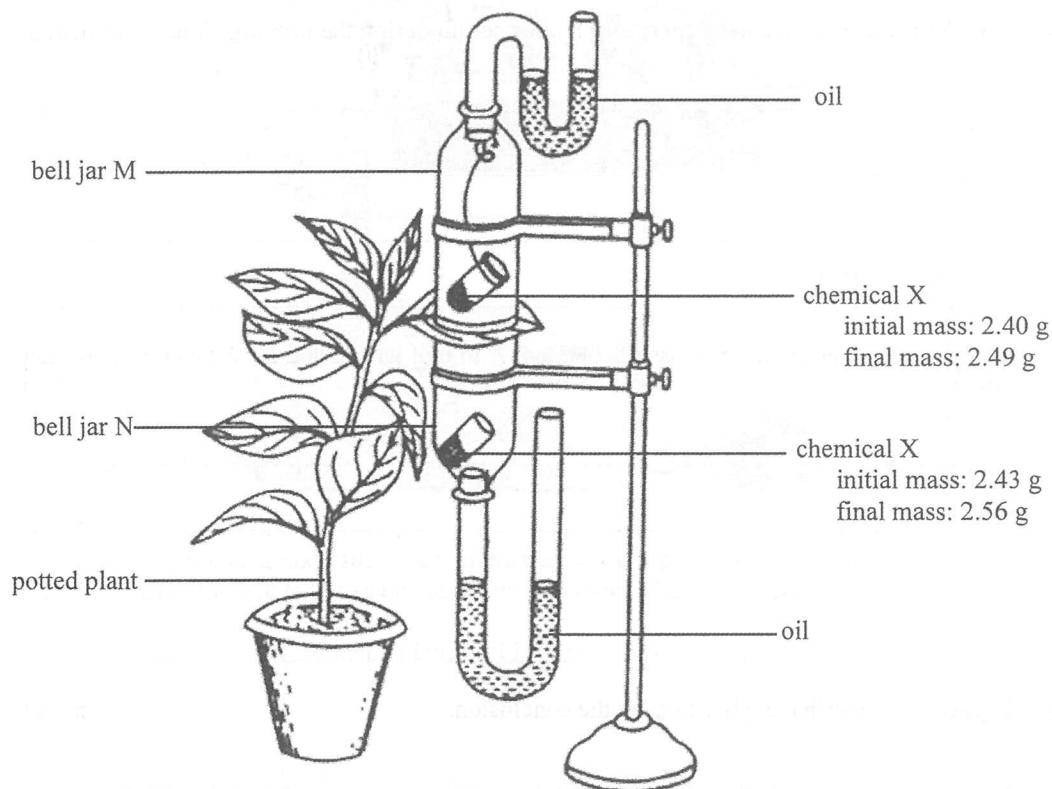
- (ii) Give **one** digested product which is absorbed through process Y in the human body. (1 mark)

- (iii) The digested products absorbed through process X are transported from the small intestine to the heart. Complete the following flowchart to show the vessels involved in the transport. (3 marks)



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5. The diagram below shows an experimental set-up used to compare the transpiration rates of the upper and lower epidermis of the leaf of a potted plant. The set-up consists of two identical bell jars placed one above the other with the leaf of a potted plant in between. Chemical X was placed into the jars to absorb water vapour. The whole set-up was made air-tight. The masses of the chemical X in the jars were measured at the beginning and after five hours.



- (a) (i) Which of the following parameters is used to measure the transpiration rate of the plant in this investigation? Put a '✓' in the appropriate box to indicate your choice. (1 mark)

- ☐ amount of water lost by the leaf
☐ amount of water absorbed by the leaf

- (ii) How can the above set-up measure the parameter chosen in (a) (i)? (2 marks)

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- (b) List **two** variables which have been controlled by this set-up during the investigation. (2 marks)

- (c) (i) Compare the changes in the mass of chemical X in bell jars M and N. What conclusion can you draw? (2 marks)

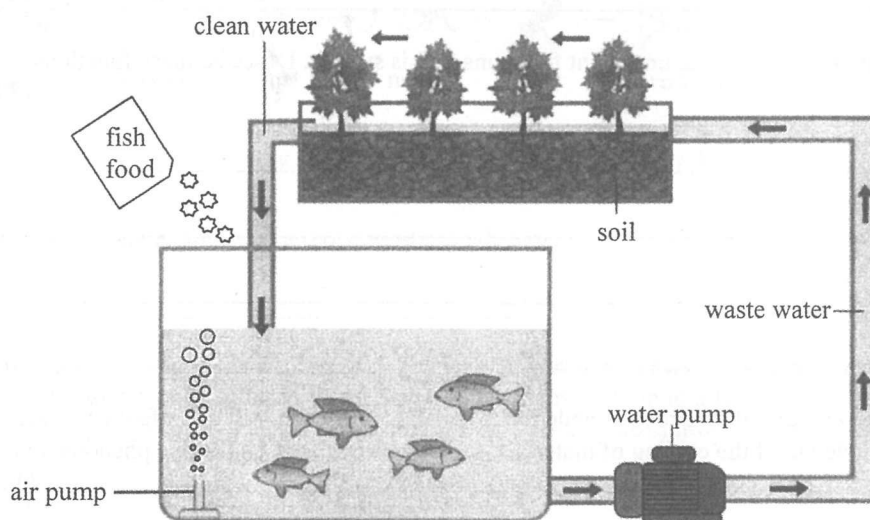
- (ii) Suggest **one** possible explanation for the conclusion. (1 mark)

- (iii) Propose **one** workable method to test your hypothesis in (c) (ii). (2 marks)

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6. The diagram below shows a mini-ecosystem in which waste water from a fish tank is used as a source of nutrients for plant growth by making use of the interaction among fish, microorganisms and plants. Adding fertilisers and periodic change of water are not necessary. This is an eco-friendly way to grow vegetables and raise fish for human consumption.



- (a) Ammonia, a toxic substance, is the major waste product excreted by fish. Ammonia in waste water from the fish tank can be converted to nitrate, which is required by plants for growth.

- (i) Name the bacteria involved in the conversion. (1 mark)

- (ii) Describe how plants can obtain nitrate from waste water and make use of it for protein synthesis in their leaves. (3 marks)

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- (b) The air pump performs some important functions in this system. Describe these functions. (2 marks)

(1)

(2)

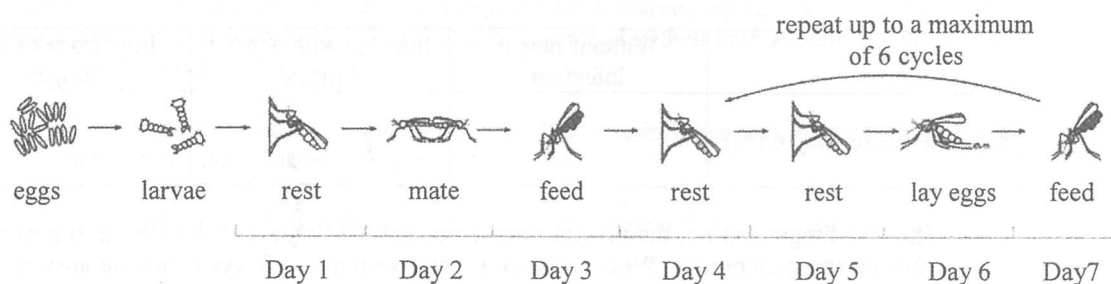
- (c) If double the amount of fish food is added accidentally, some fish will die a few days later. Based on your knowledge of the cycling of materials, suggest an explanation for this phenomenon. (4 marks)

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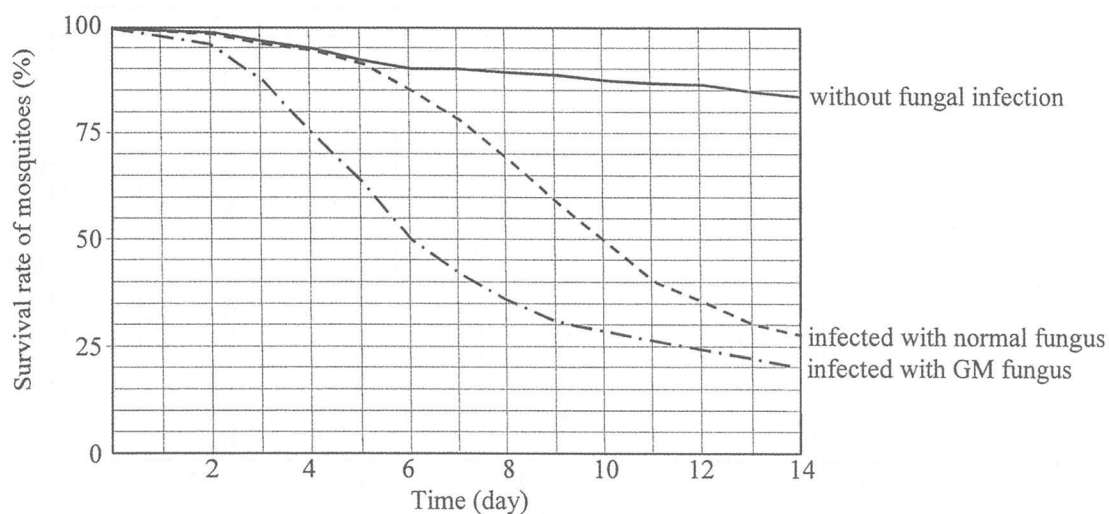
7. The diagram below shows the life cycle of mosquitoes:



- (a) Give **one** example of a mosquito-borne disease. (1 mark)

- (b) The Food and Environmental Hygiene Department introduced a fungal species to control the mosquito population in Hong Kong. Mosquitoes will die in a few days if they are infected with the fungus. What is the ecological relationship between mosquitoes and this fungal species? (1 mark)

- (c) To make this fungal species more deadly to mosquitoes, researchers produced a genetically-modified (GM) fungus which can produce a toxin in the mosquitoes. The effectiveness of this biological control was examined by measuring the survival rate of adult mosquitoes over 14 days. The results are shown below:



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- (i) Using the data from the graph, complete the following table. (1 mark)

	Without fungal infection	Infected with normal fungus	Infected with GM fungus
Survival rate on Day 6 (%)			

- (ii) The GM fungus but not the normal fungus was able to 'wipe out' the mosquito population after several generations. With reference to the mosquito's life cycle and the answer in (c) (i), explain this phenomenon. (3 marks)

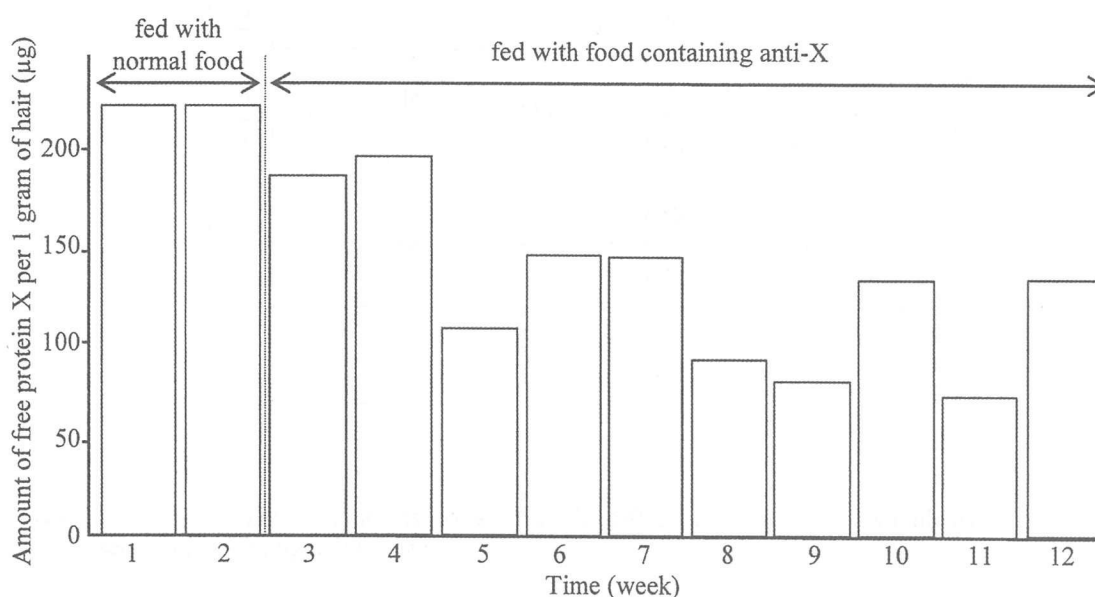
8. Some people suffer from sneezing and coughing when there are cats nearby. These unwanted immune responses, known as allergies, are caused by a protein X secreted by cats' salivary and sebaceous glands. When cats lick their body surface, this protein X is spread to their hair. Protein X can accumulate in their living environment over time.

(a) Recent research shows that the amount of free protein X on cats' hair can be reduced by adding antibodies against protein X (anti-X) to cat food.

(i) Why can anti-X reduce the amount of free protein X on cats' hair? (1 mark)

(ii) There are no proteases in the saliva of cats. Explain why this is important for the success of this method of reducing free protein X. (1 mark)

(b) In the research, a group of domestic cats were fed with normal cat food for 2 weeks, followed by cat food containing anti-X for 10 weeks. The amount of free protein X found on their hair during the research is shown in the bar chart below:



(i) Research team member A thought anti-X was effective in reducing the amount of free protein X on cats' hair while team member B thought anti-X was not effective. Based on the bar chart, give **one** reason to support team member A and **another** reason to support team member B. (2 marks)

Supporting reason for team member A:

Supporting reason for team member B:

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- (ii) How would you modify the research to confirm whether anti-X in cat food is effective in reducing the amount of free protein X on cats' hair? (2 marks)

- (c) Based on the information in the introductory paragraph of this question, suggest **two** limitations of this approach to reducing allergies caused by cats. (2 marks)



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9. The following DNA sequence shows the coding strand of part of a gene found in insect species A:

... ATG GTC GTA TAC GCT ACC CTG TCG ATG CTA GCT AGC ...

- (a) Which of the following correctly shows the sequence of the mRNA corresponding to the underlined sequence of this coding strand? Put a '✓' in the appropriate box to indicate your choice. (1 mark)

☐ AUG GUC GUA UAC GCU ACC

☐ UAC CAG CAU AUG CGA UGG

- (b) Using the following codon table, write the amino acid sequence of the protein translated from the mRNA in (a). (2 marks)

		Second base of the codon					
		U	C	A	G		
First base of the codon	U	Phe	Ser	Tyr	Cys	U	Third base of the codon
		Leu		STOP	STOP Trp	C A G	
	C	Leu	Pro	His	Arg	U C A G	
				Gln		U C A G	
	A	Ile	Thr	Asn	Ser	U C A G	
		Met		Lys	Arg	U C A G	
	G	Val	Ala	Asp	Gly	U C A G	
				Glu		U C A G	

- (c) It was found that the gene has two alleles. The difference between the two alleles in the underlined sequence is highlighted below:

allele1: ... ATG GTC GTA TA C GCT ACC ...

allele 2: ... ATG GTC GTA TA G GCT ACC ...
(mutated)

- (i) What kind of mutation is this? (1 mark)

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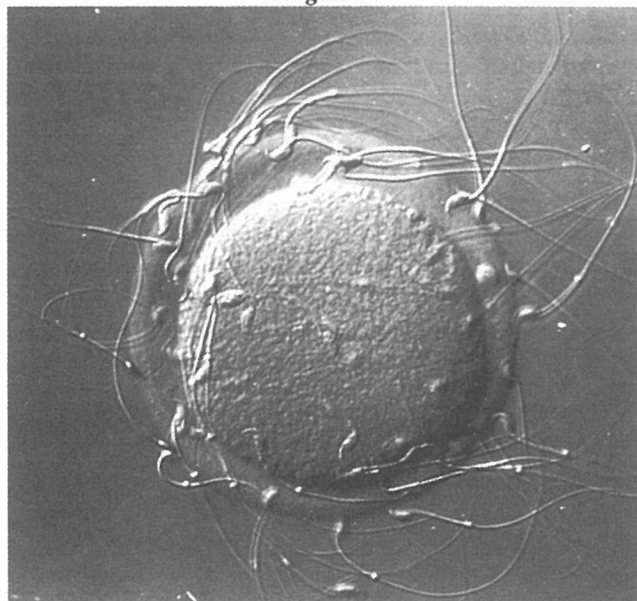
- (ii) With reference to the codon table, describe how this mutation affects the protein translated from this gene. (3 marks)

- (iii) Individuals from insect species A exist in two forms, green bodied and brown bodied. Individuals with a green body have only allele 1. Some of those with a brown body have both allele 1 and allele 2 while others have only allele 2.

If the body colour of insect species A was only caused by the mutation of this gene, which allele (1 or 2) would be recessive? Explain your answer. (3 marks)
(Note: Marks will not be awarded for genetic diagrams.)

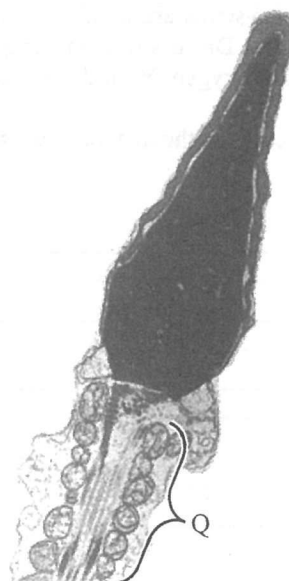
10. Diagram I shows a photomicrograph of human sperm and a human ovum during fertilisation while Diagram II shows an electron micrograph of an enlarged view of human sperm.

Diagram I



(700X)

Diagram II



(30 000X)

- (a) Under normal circumstances, in which structure of the female reproductive system does the process shown in Diagram I take place? (1 mark)

- (b) With reference to Diagram II, what is the significance of organelles Q to the sperm's function? (1 mark)

- (c) (i) Explain the significance of the chromosome number of the sperm and ovum to sexual reproduction. (2 marks)

- (ii) Briefly describe how identical twins may arise after fertilisation. (2 marks)

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You are required to present your answer to the following question in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Our body systems are highly specialised at various levels of organisation to facilitate the functioning of each system. Discuss how the adaptive features of the human blood and circulatory system facilitate the transport of oxygen. Your discussion should cover adaptive features at cellular, tissue and organ levels.

(11 marks)

(Note: Details of the different types of blood vessels are not required.)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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